



IBM Quantum



OFFICIAL EVENT

Qiskit

Fall Fest

2026

Circuit Optimization & Transpilation



Qiskit Fall Fest 2026 at University of Engineering & Technology Lahore, Narowal Campus

Review

What have we learned so far?

- **Qubits**
 - State vectors, Bra-ket notation, Basis states
- **Measurement**
 - Probability amplitudes, Born rule
- **Multiple Qubits**
 - Tensor product, Product states, Entangled states
- **Operators**
 - Unitary matrices, Hermitian matrices
- **Circuits**



Qubits

States as Vectors



Qubits are often represented with a **column vector** called a **ket**. Each ket has an associated **row vector** called a **bra**.

$$|\psi\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix} \quad \text{and} \quad \langle\psi| = [\alpha^* \quad \beta^*]$$

α and β are **complex numbers**, and α^* and β^* are their **complex conjugates**.

Complex conjugation means that if $\alpha = x + iy$, then $\alpha^* = x - iy$.

Importantly, α and β are **normalized**: $|\alpha|^2 + |\beta|^2 = 1$

Qubits

Vectors Need a Basis



We always choose a **basis** to work in. The *standard* or *computational basis states* in quantum computing are:

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \text{and} \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

A qubit in a **pure state** can be written in several equivalent ways:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle = \alpha \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

This **linear combination** of basis states is called **superposition**.

Measurement

Coefficients are Probability Amplitudes



By the *Born rule*, probabilities are the squared magnitudes of the amplitudes.

The *probability amplitudes* for $|0\rangle$ and $|1\rangle$ are the coefficients α and β , so:

$$P(0) = |\alpha|^2 \quad \text{and} \quad P(1) = |\beta|^2$$

This is why the coefficients must be normalized.

$$|\alpha|^2 + |\beta|^2 = 1$$

Note, if $\alpha = x + iy$ then:

$$|\alpha|^2 = \alpha^* \alpha = (x - iy)(x + iy) = x^2 + y^2$$

Measurement

Finding an Outcome



Measuring in the computational basis gives either 0 or 1.

$$|\psi\rangle = \frac{\sqrt{3}}{2} |0\rangle + \frac{1}{2} |1\rangle$$

Applying the Born rule to $|\psi\rangle$, the probabilities of measuring either 0 or 1 are:

$$P(0) = \left| \frac{\sqrt{3}}{2} \right|^2 = \frac{3}{4} \qquad P(1) = \left| \frac{1}{2} \right|^2 = \frac{1}{4}$$

After measurement, the qubit's state becomes the basis state of the measured outcome.

$$0 \rightarrow |0\rangle \qquad 1 \rightarrow |1\rangle$$

Multiple Qubits

Two Qubits, Four Basis States



Two qubits have four associated basis states:

$$|00\rangle, |01\rangle, |10\rangle, |11\rangle$$

The associated state vector has four terms:

$$|\psi\rangle = a|00\rangle + b|01\rangle + c|10\rangle + d|11\rangle$$

Amplitudes determine the probabilities of each result, so they must be normalized:

$$|a|^2 + |b|^2 + |c|^2 + |d|^2 = 1$$

Multiple Qubits

Combining States



The *tensor product* (also known as the *Kronecker product*) combines the states of multiple qubits.

$$|0\rangle \otimes |1\rangle = |01\rangle$$

$$|1\rangle \otimes |0\rangle = |10\rangle$$

Generally:

$$\begin{bmatrix} a \\ b \end{bmatrix} \otimes \begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} ac \\ ad \\ bc \\ bd \end{bmatrix}$$

Multiple Qubits

States We Can Separate



If a multi-qubit state can be written as a tensor product of individual pure states it is called a *product state*.

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \rightarrow |+\rangle \otimes |+\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$$

However, some states cannot be written as a product state, for example:

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

Multiple Qubits

States We Cannot Separate



When a state cannot be written as a tensor product of individual pure states it is called an *entangled state*.

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

A measurement of either qubit determines the outcome of a measurement of the other qubit.

For $|\Phi^+\rangle$, the only possible outcomes for a measurement in the computational basis are $|00\rangle$ and $|11\rangle$. Each with probability $1/2$.

Operators

Matrices Act on States



Operators are linear transformations that act on the vectors representing quantum states. They are represented by matrices:

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

If an operator A acts on $|\psi\rangle$, then: $A|\psi\rangle = |\psi'\rangle$

$$X|0\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad \text{and} \quad X|1\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Operators

Evolution and Measurement



Unitary operators describe reversible evolutions of quantum states.

$$U^\dagger U = U U^\dagger = I$$

Hermitian operators describe **observables**, their real eigenvalues are the possible measurement outcomes for that observable.

$$A^\dagger = A$$

Circuits

Wires and Gates

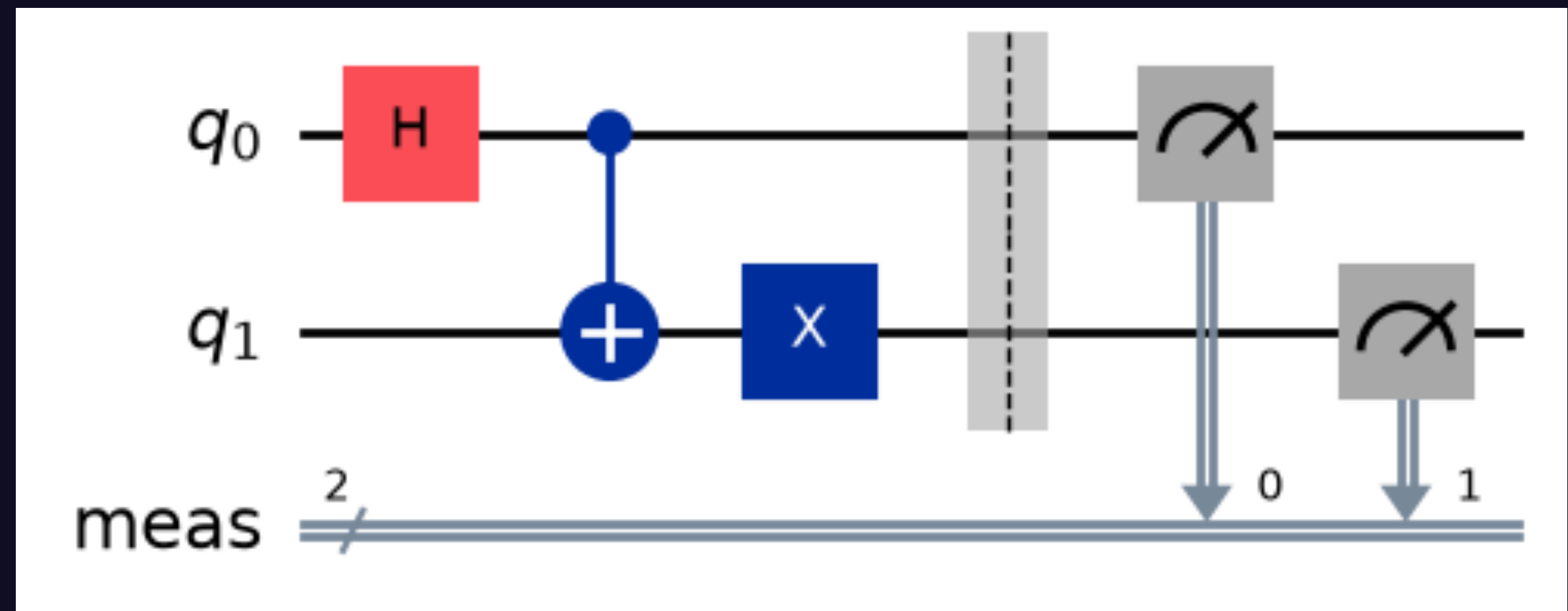


It is common to represent qubits and operators as a **circuit**.

Qubits are represented as wires, with each gate acting on them from the left to the right.

Single qubit gates: X, Y, Z, H

Multi-qubit gates: $CX, SWAP$



Topics



- **Circuit Depth**

- Gates Take Time, Why Depth Matters, Same Result Different Depth

- **Reducing Depth**

- The GHZ State, Fan-Out vs. Chain, Recursive Fan-Out

- **Transpilation**

- Hardware Has Rules, The Coupling Map, What the Transpiler Does, Transpilation Can Increase Depth

- **Heavy-Hex Topology**

- How Heron Qubits Connect, GHZ on Heavy-Hex

Circuit Depth

Gates Take Time



Gates represent operations being done on qubits, but they must be done in the correct sequence and each operation takes time to complete.

Operations on different qubits can often happen at the same time.

Operations that depend on each other or act on the same qubit must happen sequentially.

Circuit Depth

Why Depth Matters



So far we have discussed *abstract circuits*. These are idealized with perfect qubits, perfect operators, and complete connectivity between qubits.

A qubit in a real quantum computer is a physical object.

The qubit's environment can disturb it, the equipment applying operators to it is imperfect, and normally not all qubits in the quantum computer can communicate directly with each other.

A circuit's *depth* is the maximum number of sequential operations it requires.

"Shallow" circuits, with less depth, generally have less opportunity for these errors to accumulate.

Circuit Depth

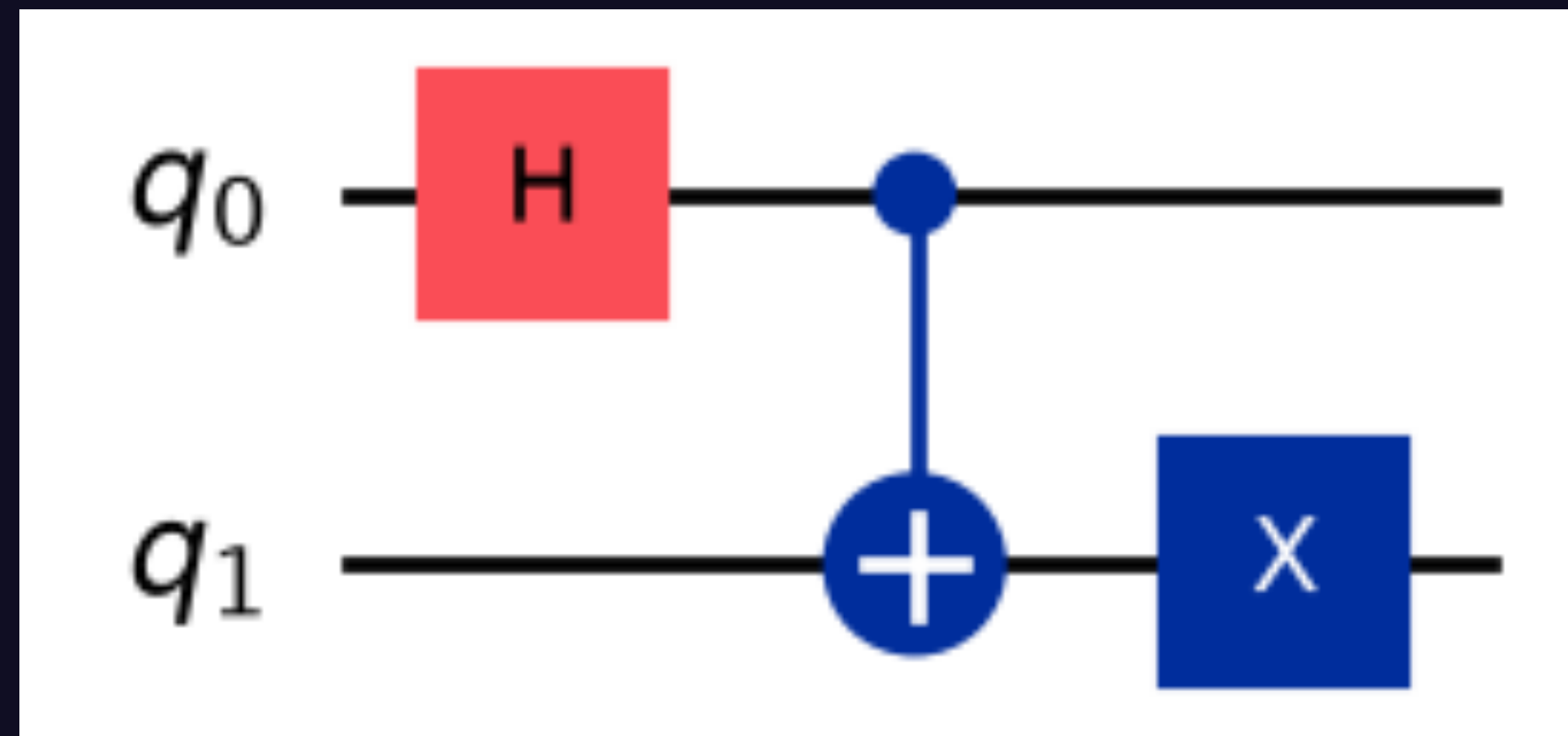
Same Result, Different Depth



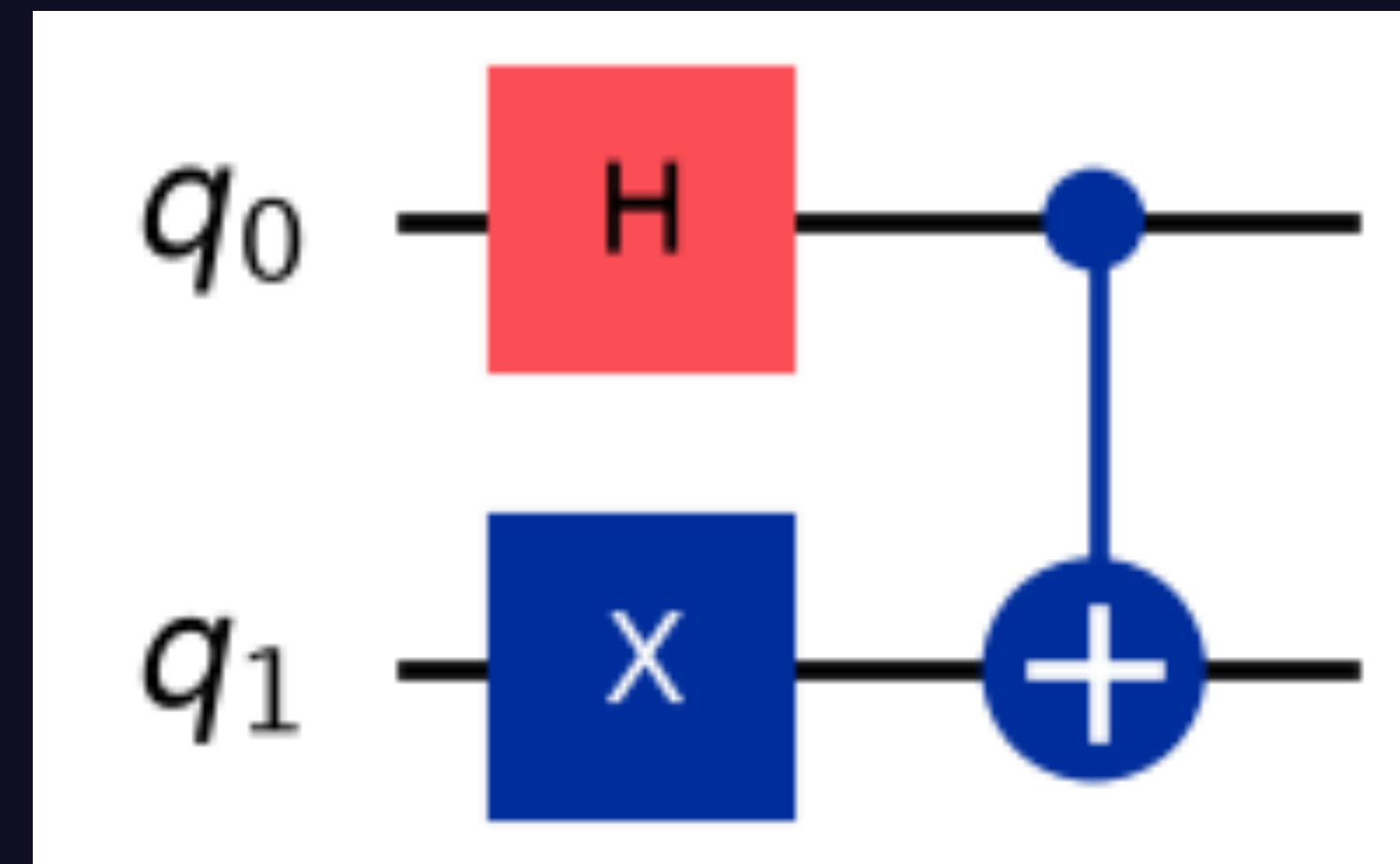
Both circuits result in the same state:

$$|\Psi^+\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}}$$

Depth = 3



Depth = 2



Reducing Depth

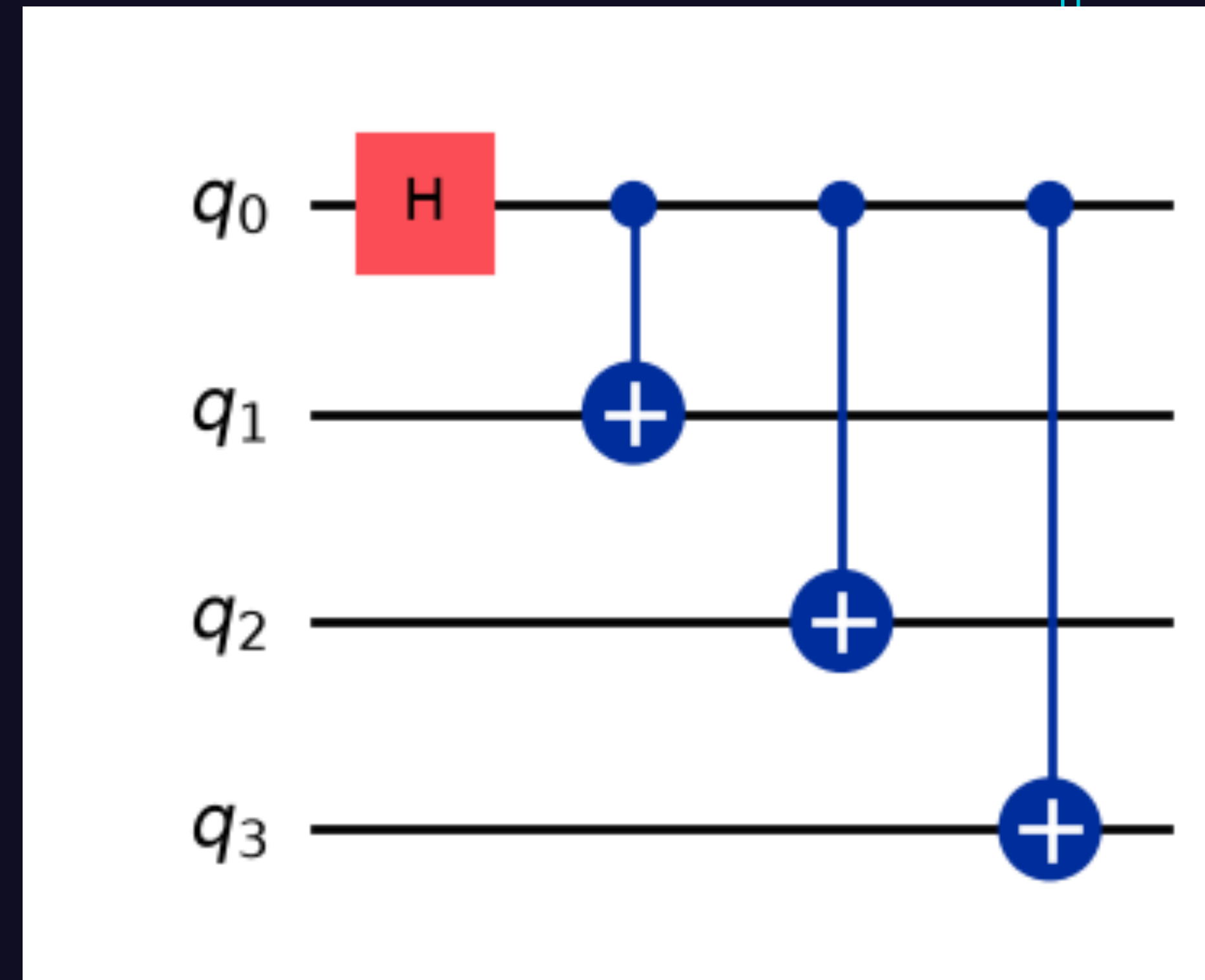
The GHZ State

This is the *fan-out* method to prepare an n -bit GHZ state: apply H to q_0 followed by $n - 1$ CX gates from q_0 to the other qubits:

$$|GHZ_n\rangle = \frac{|0\rangle^{\otimes n} + |1\rangle^{\otimes n}}{\sqrt{2}}$$

Each CX gate acts on q_0 , so they must be applied sequentially.

$$\text{Depth} = 1 + (n - 1) = n$$



Reducing Depth

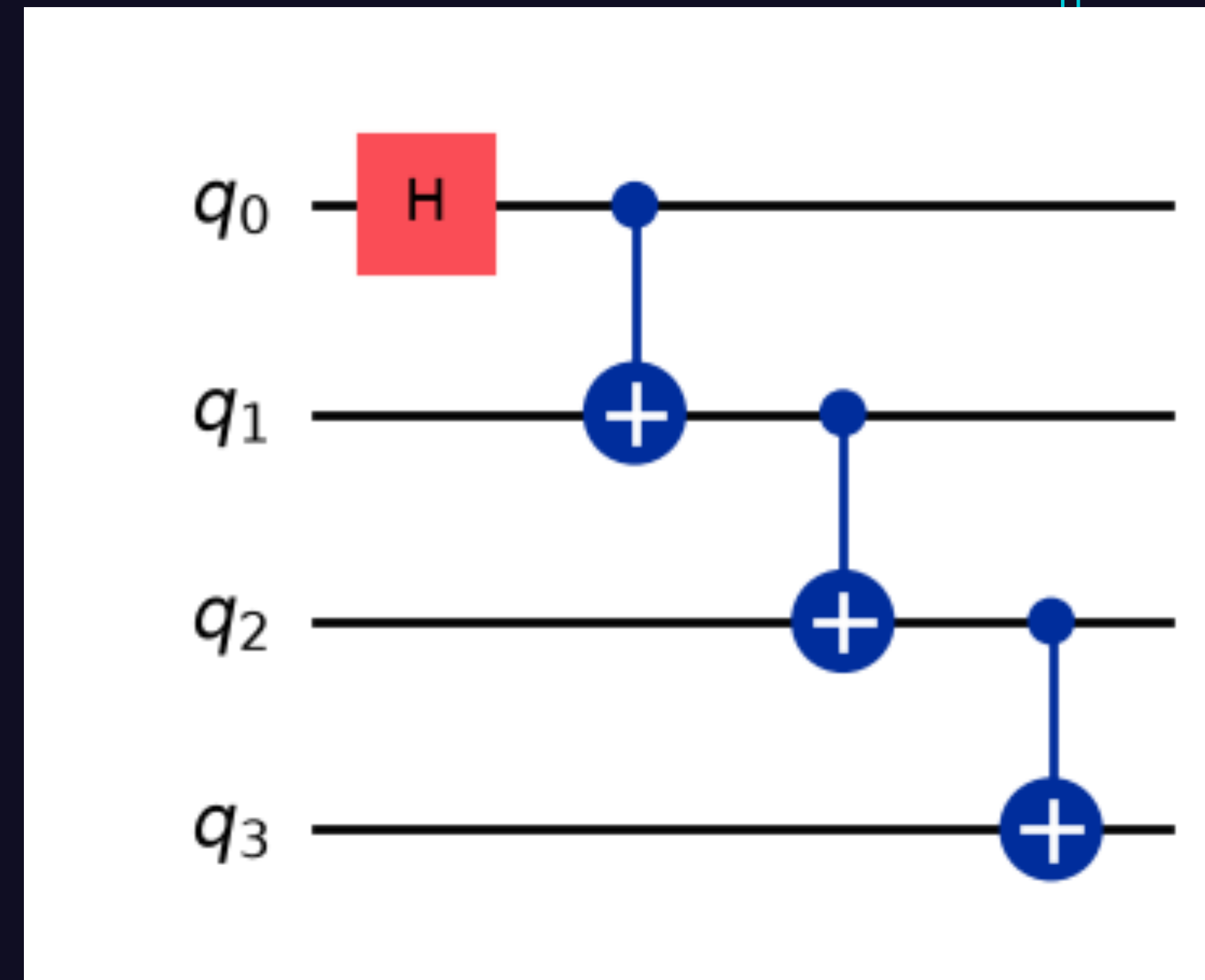
Fan-Out vs. Chain

This is the *chain* method of creating a GHZ state.

Each CX gate has a different control qubit, but they still act on qubits used by a previous gate.

$$\text{Depth} = 1 + (n - 1) = n$$

Once two qubits are entangled, either qubit can act as the control qubit of a later CX gate.



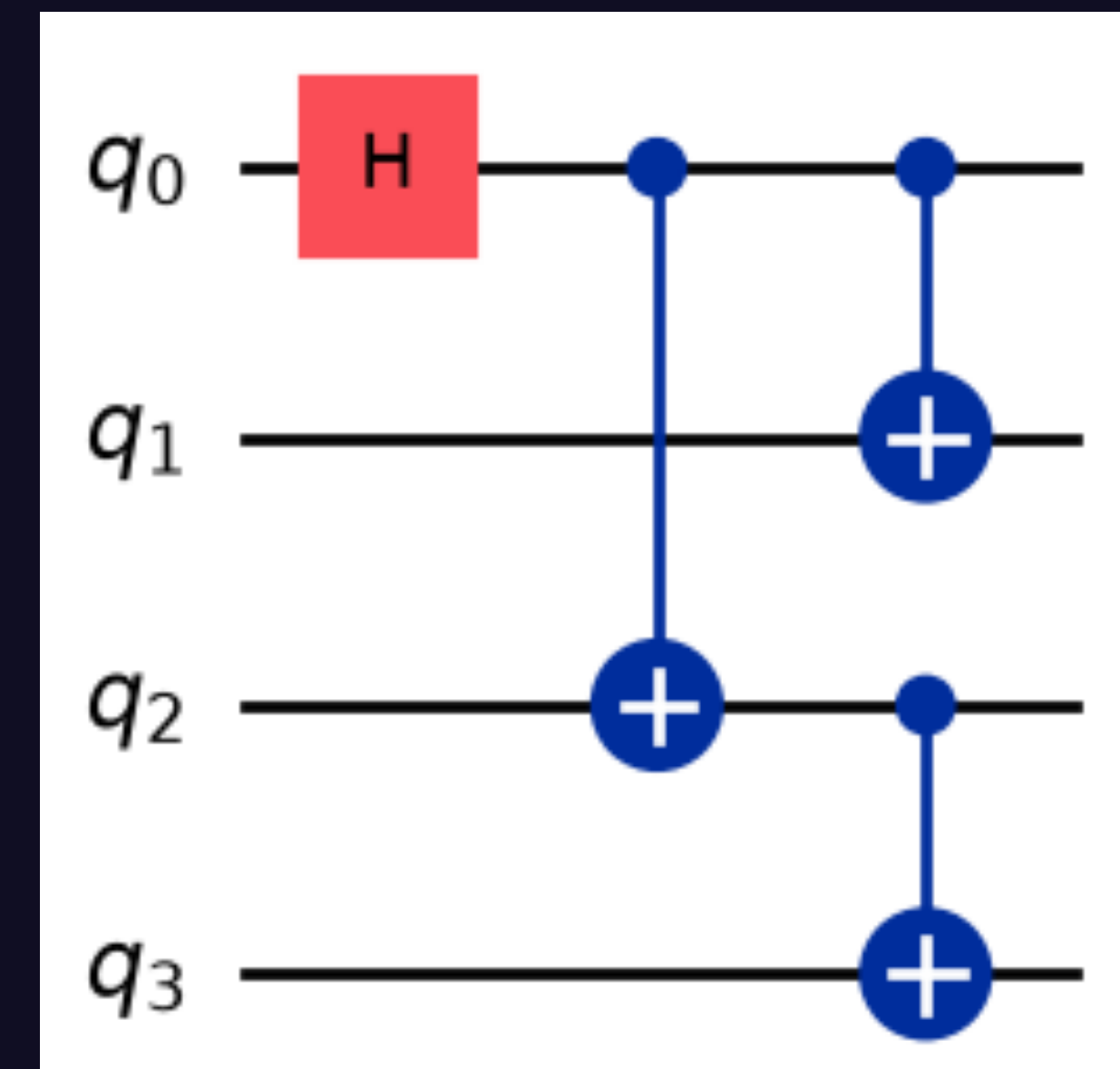
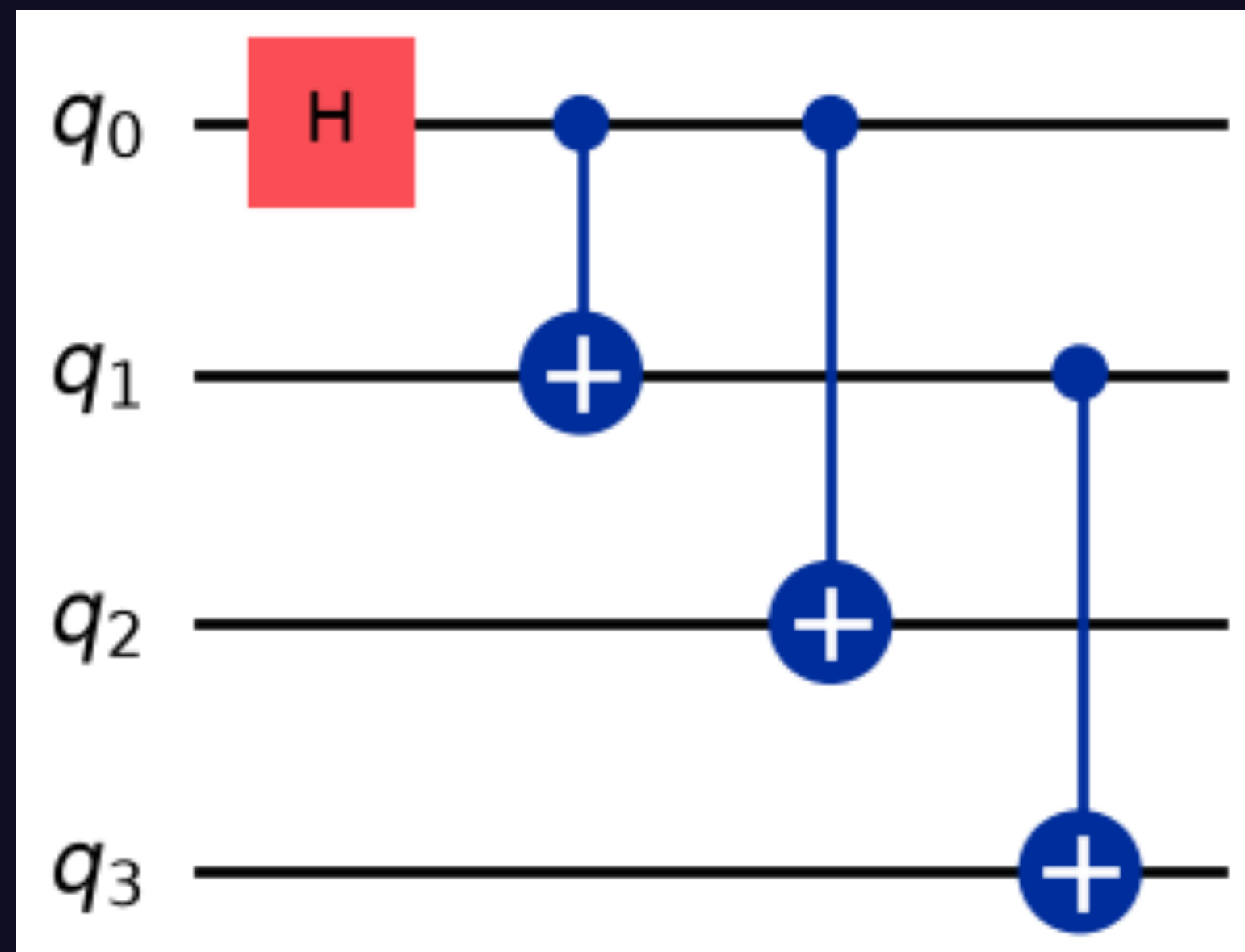
Reducing Depth

Recursive Fan-Out



Recursive fan-out uses all entangled qubits as control qubits in every layer.

$$\text{Depth} = 1 + \lceil \log_2(n) \rceil$$



Transpilation

Hardware Has Rules



So far we have optimized an abstract circuit for the GHZ state.

Real hardware imposes constraints on which qubits can directly interact and what gates are natively supported.

Transpilation is the conversion of an abstract circuit into one that can be run on a specific quantum computer.

The result of transpilation is often called an **ISA circuit**, referring to the target quantum computer's **Instruction Set Architecture**.

Transpilation

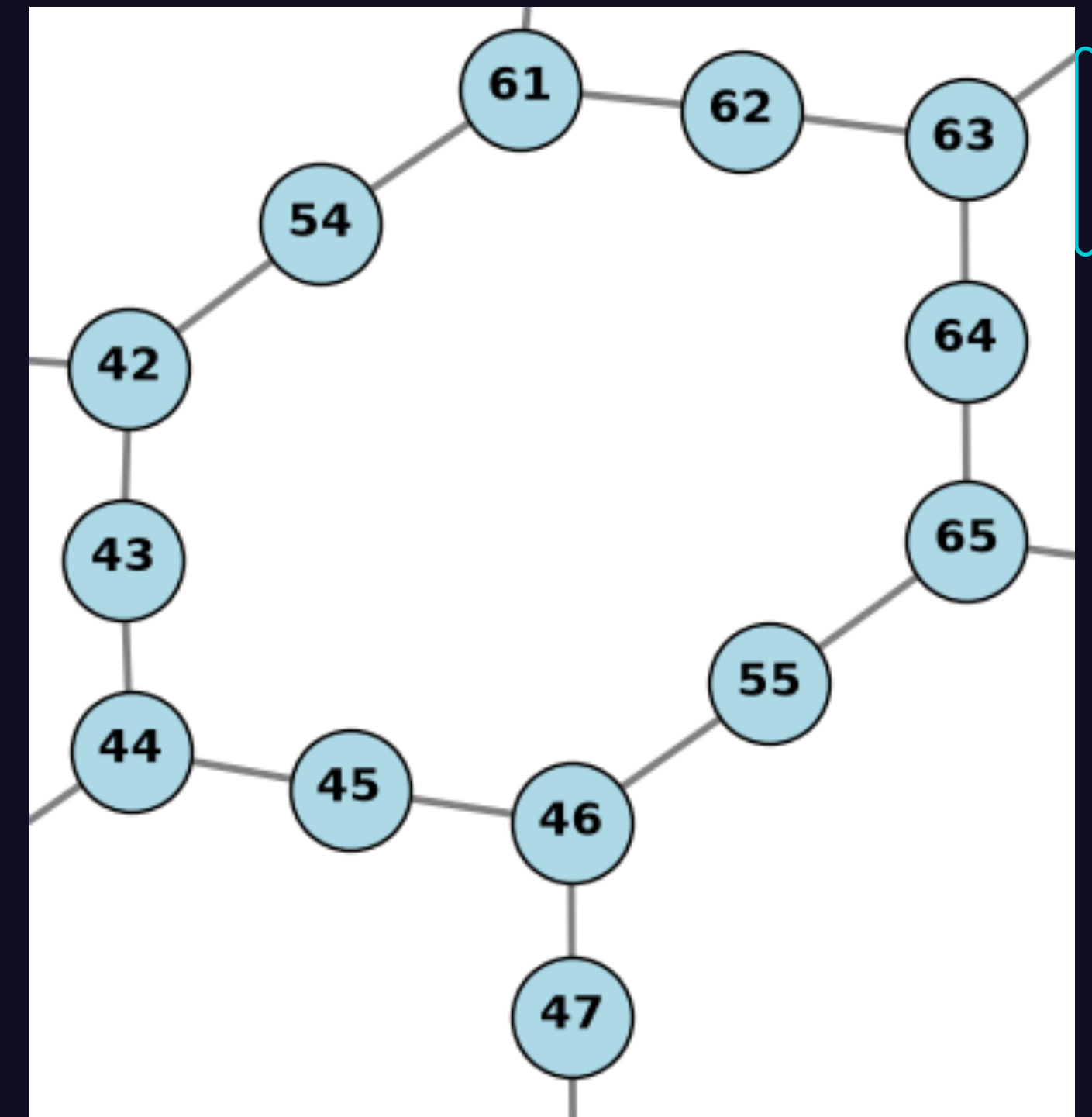
The Coupling Map

Nodes on the coupling map are physical qubits, and the edges are the physical couplers between them.

Numbers on the nodes are the physical qubit indices, and may not match the abstract circuit's qubit indices.

Two-qubits gates (like CX) cannot be made between physical qubits without coupling map connections.

If two logical qubits need to interact the transpiler must either map them to physically connected qubits in the initial layout or use $SWAP$ gates during routing to move their states together.



Transpilation

What the Transpiler Does



The Qiskit transpiler by default has six stages:

Init → *Layout* → *Routing* → *Translation* → *Optimization* → *Scheduling*

Layout maps the abstract circuit's qubits to physical qubits.

Routing identifies where *SWAPs* need to be inserted to move states to qubits that are physically connected, so that two-qubit gates can be run on them.

Translation converts the abstract circuit's gates to the native gate set of the target quantum computer.

Optimization attempts to simplify the circuit.

Transpilation

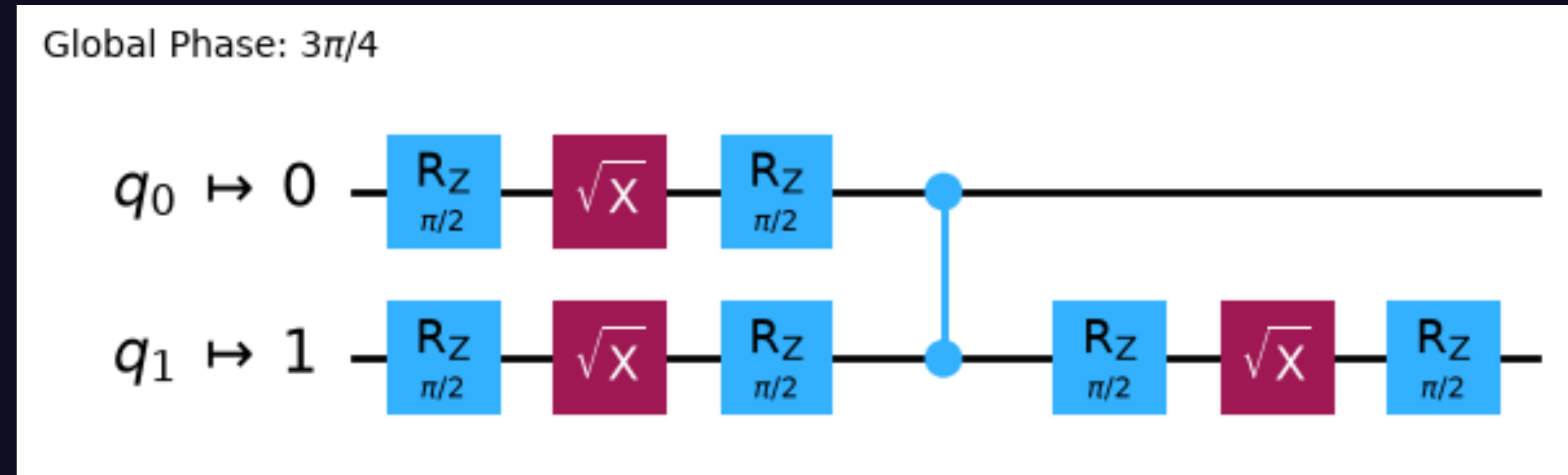
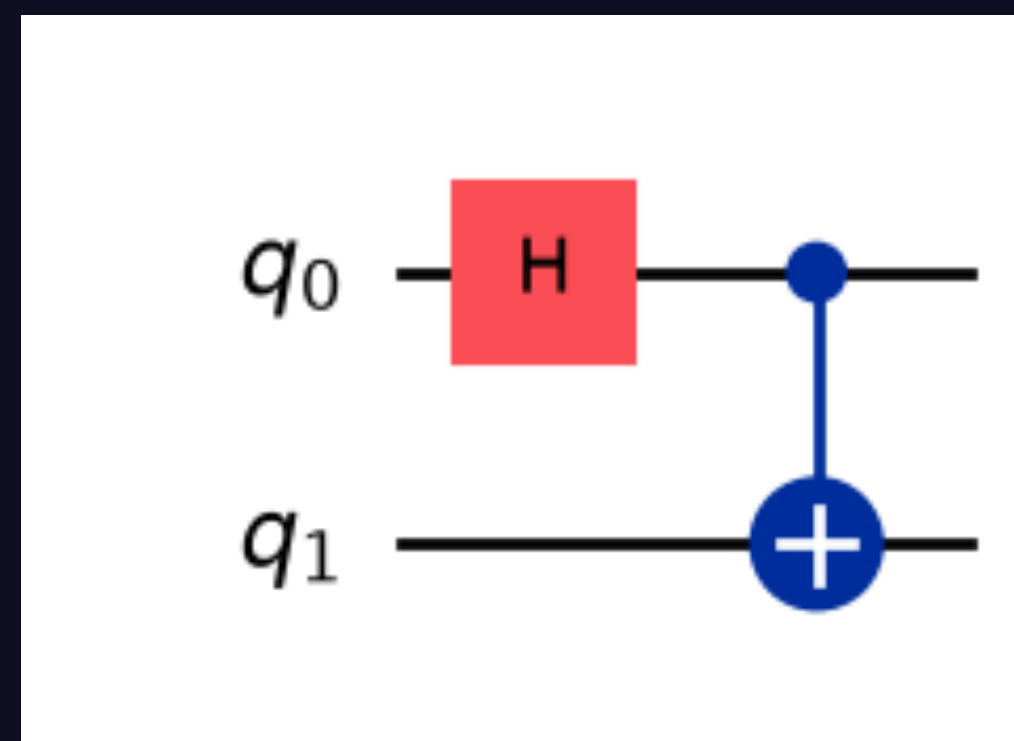
Transpilation Can Increase Depth



Universal gate sets can be used to implement any quantum computation.

Quantum processors natively implement a small *basis gate set*. All other gates must be translated into combinations of those gates.

Heron QPU basis gates: CZ, RX, RZ, SX, X



Heavy-Hex Topology

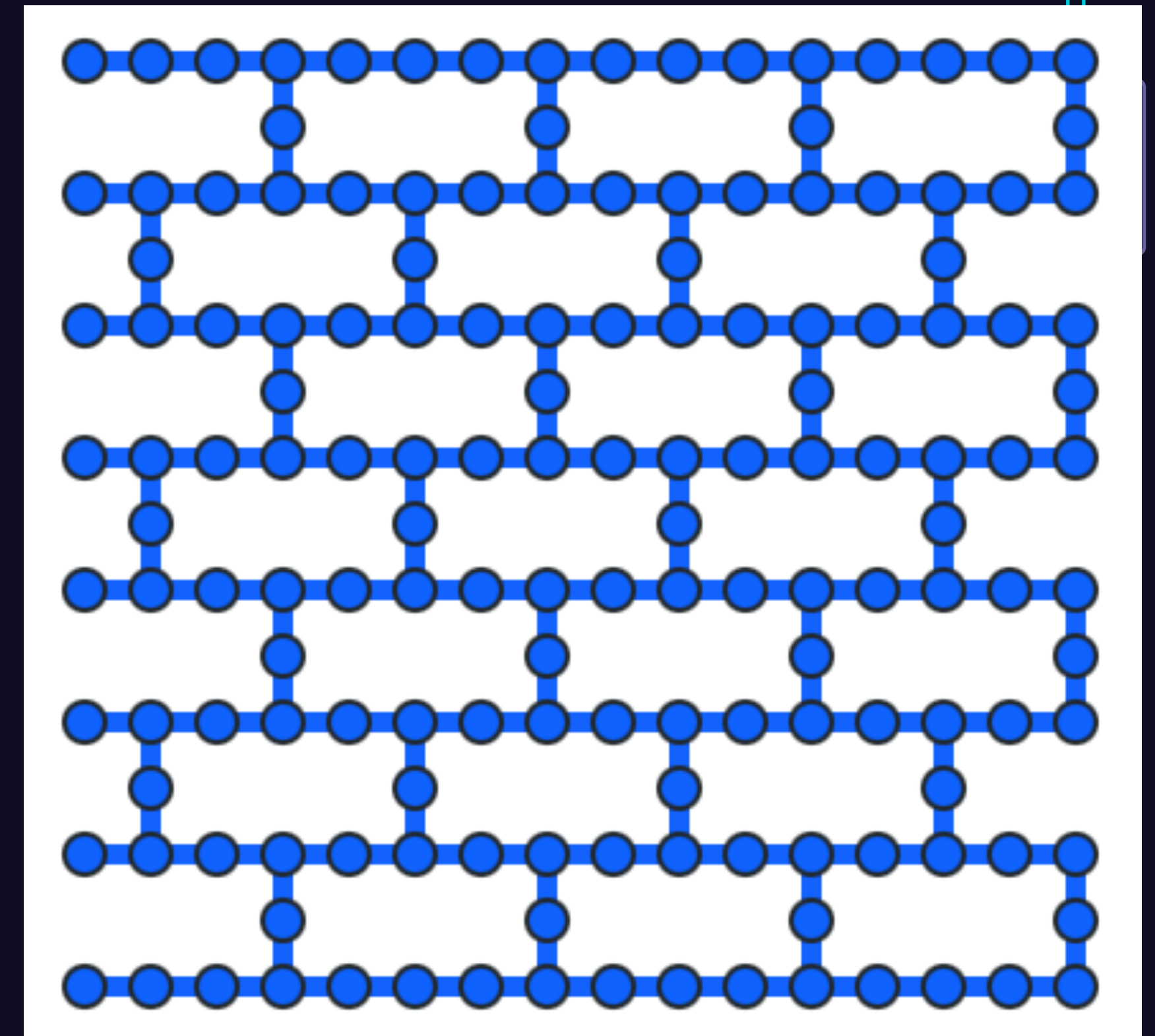
How Heron Qubits Connect

Heavy-hex is the topology used on Heron QPUs.

Most qubits are connected directly to 2 or 3 others.

This simplifies hardware engineering, but restricts circuit layout.

This means that good layout and routing during transpilation is very important.



Heavy-Hex Topology

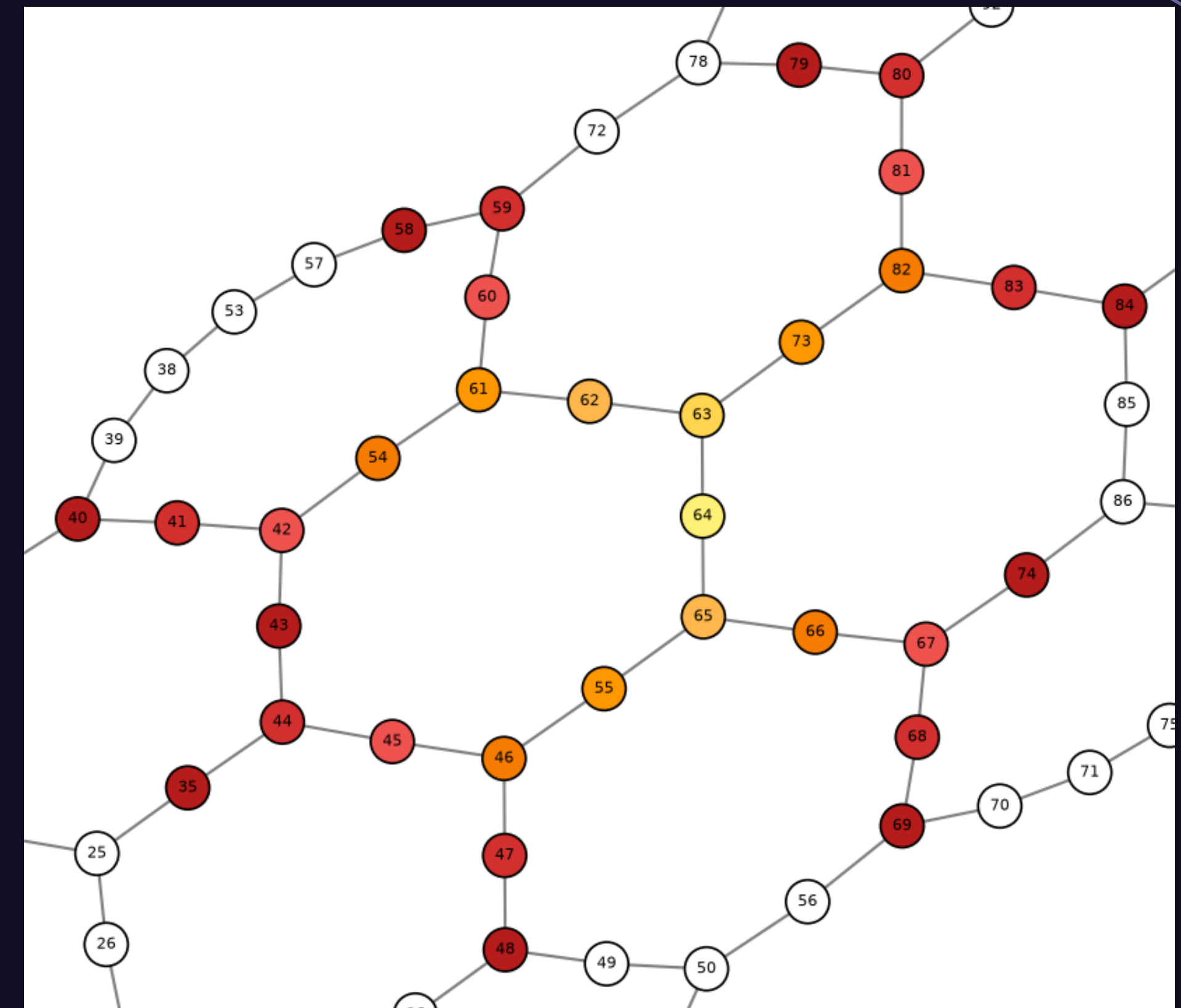
GHZ on Heavy-Hex

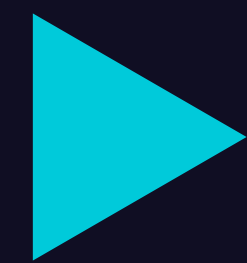


The abstract recursive fan-out GHZ circuit can entangle 256 qubits with 8 layers of CX gates.

Starting from physical qubit 64 on this Heavy-Hex coupling map we are only able to entangle 42 qubits with 8 layers of CX gates.

Hardware connectivity limits how quickly entanglement can be spread.





Questions?

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